

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

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JAVA ASSIGNMENT

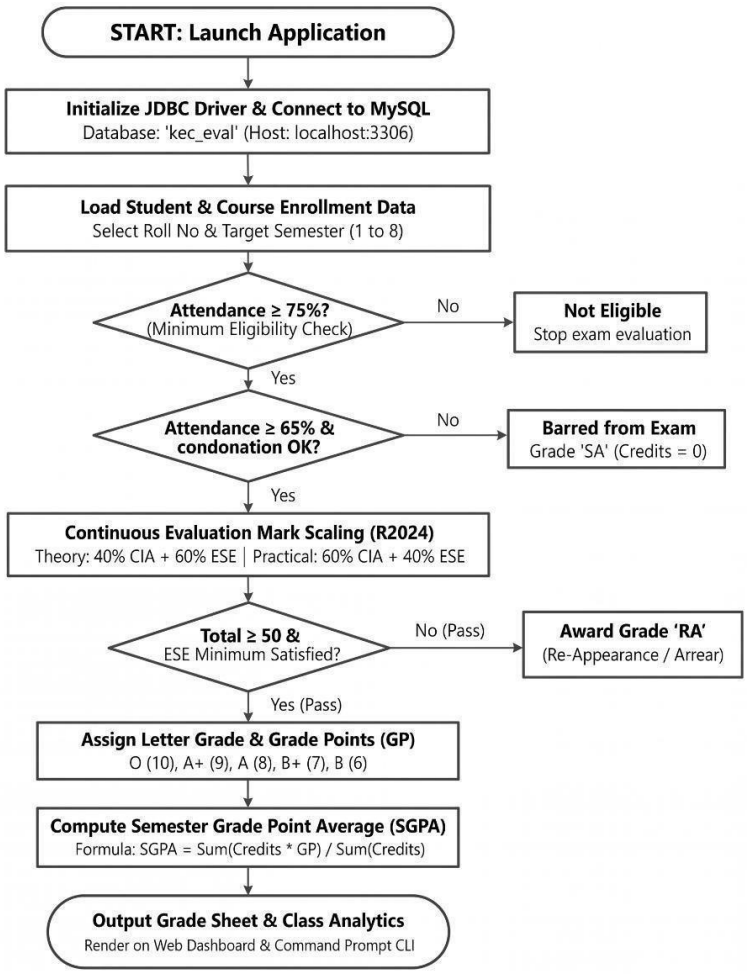
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

DATABASE SCHEMA:

1. STUDENTS TABLE:

```
mysql> DESC students;
+------------------+-------------+------+-----+---------+-------+
| Field            | Type        | Null | Key | Default | Extra |
+------------------+-------------+------+-----+---------+-------+
| roll_no          | varchar(20) | NO   | PRI | NULL    |       |
| student_name     | varchar(100)| NO   |     | NULL    |       |
| department       | varchar(50) | NO   |     | NULL    |       |
| current_semester | int         | NO   |     | NULL    |       |
+------------------+-------------+------+-----+---------+-------+
4 rows in set (0.01 sec)
```

2. COURSES TABLE:

```
mysql> DESC courses;
+--------------+-----------------------------------------+------+-----+---------+-------+
| Field        | Type                                    | Null | Key | Default | Extra |
+--------------+-----------------------------------------+------+-----+---------+-------+
| course_code  | varchar(20)                             | NO   | PRI | NULL    |       |
| course_title | varchar(100)                            | NO   |     | NULL    |       |
| credits      | int                                     | NO   |     | NULL    |       |
| course_type  | enum('THEORY','PRACTICAL','EMBEDDED')   | NO   |     | NULL    |       |
+--------------+-----------------------------------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

3. ENROLLMENTS TABLE:

```
mysql> DESC enrollments;
+--------------------+-------------+------+-----+---------+----------------+
| Field              | Type        | Null | Key | Default | Extra          |
+--------------------+-------------+------+-----+---------+----------------+
| enrollment_id      | int         | NO   | PRI | NULL    | auto_increment |
| roll_no            | varchar(20) | NO   | MUL | NULL    |                |
| course_code        | varchar(20) | NO   | MUL | NULL    |                |
| semester           | int         | NO   |     | NULL    |                |
| classes_held       | int         | YES  |     | 0       |                |
| classes_attended   | int         | YES  |     | 0       |                |
| medical_condonation| tinyint(1)  | YES  |     | 0       |                |
+--------------------+-------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

4.ASSESSMENT MARKS TABLE:

```
mysql> DESC assessment_marks;
+----------------+--------------+------+-----+---------+-------+
| Field          | Type         | Null | Key | Default | Extra |
+----------------+--------------+------+-----+---------+-------+
| enrollment_id  | int          | NO   | PRI | NULL    |       |
| cia1           | decimal(5,2) | YES  |     | NULL    |       |
| cia2           | decimal(5,2) | YES  |     | NULL    |       |
| assignments    | decimal(5,2) | YES  |     | NULL    |       |
| quizzes        | decimal(5,2) | YES  |     | NULL    |       |
| lab_evaluation | decimal(5,2) | YES  |     | NULL    |       |
| ese_marks      | decimal(5,2) | YES  |     | NULL    |       |
| ese_absent     | tinyint(1)   | YES  |     | 0       |       |
+----------------+--------------+------+-----+---------+-------+
8 rows in set (0.00 sec)
```

SOURCE CODE:

MAINAPP.JAVA:

```
package kec;

import kec.dao.StudentDAO;
import kec.dao.AcademicDAO;
import java.sql.SQLException;
import java.util.Scanner;

public class MainApp {     private static final Scanner IN = new
Scanner(System.in);     private static final StudentDAO STUDENTS =
new StudentDAO();     private static final AcademicDAO ACADEMICS =
new AcademicDAO();

   private static String ask(String prompt) { System.out.print(prompt); return IN.nextLine().trim(); }
private static int askInt(String prompt) { return Integer.parseInt(ask(prompt)); }     private static
double askDouble(String prompt) { return Double.parseDouble(ask(prompt)); }     private static
boolean yes(String prompt) { return ask(prompt + " (y/n): ").equalsIgnoreCase("y"); }

   public static void main(String[] args) {
      System.out.println("KEC R2024 Student Academic & Continuous Evaluation System");
while (true) {
         System.out.println("\n1 Add student  2 Add course  3 Enroll student + attendance");
         System.out.println("4 Update attendance  5 Enter/update marks  6 Student mark sheet");
         System.out.println("7 Attendance defaulters  8 Department result summary  0 Exit");
          String choice = ask("Choose: ");
try {             switch (choice) {
case "1":
```

```
                STUDENTS.addStudent(ask("Roll number: ").toUpperCase(), ask("Student name: "),
ask("Department: ").toUpperCase(), askInt("Current semester (1-8): "));
System.out.println("Student added.");
                 break;
case "2":
                STUDENTS.addCourse(ask("Course code: ").toUpperCase(), ask("Course title: "),
askDouble("Credits: "), ask("Type (THEORY/PRACTICAL/EMBEDDED): "));
                System.out.println("Course added.");
break;                  case "3": {
                String roll = ask("Roll number: ").toUpperCase(), course = ask("Course code: ").toUpperCase();
int sem = askInt("Semester: "), held = askInt("Classes held: "), attended = askInt("Classes attended: ");
                boolean cond = yes("Medical condonation approved");
                long id = STUDENTS.enroll(roll, course, sem, held, attended, cond);
System.out.println("Enrollment created. Enrollment ID: " + id);
break;
              }
case "4": {
                long id = Long.parseLong(ask("Enrollment ID: "));
                int held = askInt("Classes held: "), attended = askInt("Classes attended: ");
                boolean cond = yes("Medical condonation approved");
STUDENTS.updateAttendance(id, held, attended, cond);
                System.out.println("Attendance updated.");
break;
              }                 case "5": {                   long id =
Long.parseLong(ask("Enrollment ID: "));
                String field = ask("Assessment (CIA1/CIA2/ASSIGNMENT/QUIZ/LAB/ESE):
").toUpperCase();
                boolean absent = field.equals("ESE") && yes("Student absent in ESE");
                Double mark = absent ? null : askDouble("Marks (0-100): ");
                ACADEMICS.saveMarks(id, field, mark, absent);
                System.out.println("Marks saved. You can repeat this option as assessments occur.");
break;
              }
case "6":
                ACADEMICS.printMarkSheet(ask("Roll number: ").toUpperCase(), askInt("Semester: "));
break;
              case "7": ACADEMICS.attendanceDefaulters(); break;
case "8": ACADEMICS.classSummary(askInt("Semester: ")); break;
              case "0": System.out.println("Goodbye."); return;
default: System.out.println("Invalid menu choice.");
            }
        } catch (NumberFormatException ex) {
            System.out.println("Invalid number: " + ex.getMessage());
        } catch (IllegalArgumentException ex) {
            System.out.println("Input error: " + ex.getMessage());
        } catch (SQLException ex) {
            System.out.println("Database error: " + ex.getMessage());
        }
    }
  }
}
```

DBCONNECTION.JAVA:

```
package kec.db;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public final class DBConnection {
private static final String URL =
      System.getenv().getOrDefault("KEC_DB_URL", "jdbc:mysql://localhost:3306/kec_academic");
private static final String USER = System.getenv().getOrDefault("KEC_DB_USER", "root");     private
static final String PASSWORD = System.getenv().getOrDefault("KEC_DB_PASSWORD", "");
private DBConnection() {}

   public static Connection getConnection() throws SQLException {
return DriverManager.getConnection(URL, USER, PASSWORD);
   }
}
```

OUTPUT SCREENSHOTS:

MAIN MENU:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 1
```

1. ADDING STUDENT:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 1
Roll Number (e.g. 24ITR001): 25ITR124
Student Name: SAMRUKA KK
Department (CSE/ECE/MECH/IT): IT
Current Semester (1-8): 3
Student added successfully.
```

2. ADD COURSE:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 2
Course Code (e.g. 24ITR31): 25ITC04
Course Title: EVS
Credits: 1
Course Type (THEORY/PRACTICAL/EMBEDDED): THEORY
Course added successfully.
```

3. STUDENT ENROLLMENT:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 3
Roll Number: 25ITR144
Course Code: 24ITC45
Semester: 3
Enrollment successful. Enrollment ID = 8 (use this ID for marks/attendance).
```

4. UPDATE MARKS:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 4
Enrollment ID: 2
CIA Test 1 (0-100, blank to keep/zero): 85
CIA Test 2 (0-100, blank to keep/zero): 80
Assignment/Quiz/Lab (0-100, blank to keep/zero): 90
Absent in ESE? (Y/N): N
ESE Marks (0-100, blank to keep/zero): 94
Assessment marks saved successfully.
```

5. UPDATE ATTENDANCE:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 5
Enrollment ID: 2
Total classes held: 60
Classes attended: 54
Attendance updated: 90.00%
```

6. STUDENT MARKSHEET:

```
========== KEC STUDENT SEMESTER MARK SHEET ==========

Student: Subasree G | 25ITR144 | Dept: IT | Sem: 3
Course: 24ITC45 - DESIGN LAB (PRACTICAL), Credits: 2.0
CIA1: 0.00, CIA2: 0.00, Assignment/Quiz/Lab: 0.00, CIA Avg: 0.00
Internal weighted: 0.00, ESE raw: 0.00, ESE weighted: 0.00, Total: 0.00
Attendance: 0.00% | Grade: SA | Status: BARRED - SHORTAGE OF ATTENDANCE

Student: Subasree G | 25ITR144 | Dept: IT | Sem: 3
Course: 25ITC31 - Full Stack (THEORY), Credits: 3.0
CIA1: 85.00, CIA2: 80.00, Assignment/Quiz/Lab: 90.00, CIA Avg: 85.00
Internal weighted: 34.00, ESE raw: 94.00, ESE weighted: 56.40, Total: 90.40
Attendance: 90.00% | Grade: A+ | Status: PASS

SGPA: 5.40
Equivalent Percentage (SGPA x 10): 54.00%
==================================================
```

7. ATTENDANCE DEFAULT REPORT:

```
========== KEC STUDENT ACADEMIC SYSTEM ==========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice: 7

--- ATTENDANCE DEFAULTERS (<65%) : SA ---
25ITR144 | Subasree G | IT | Course 25ITC31 | 50.00% | SA
```

8.DEPARTMENT RESULT SUMMARY:

```
========= KEC STUDENT ACADEMIC SYSTEM =========
1. Add Student
2. Add Course
3. Enroll Student in Course
4. Add/Update Assessment Marks
5. Update Attendance
6. View Student Semester Mark Sheet
7. Attendance Defaulter Report
8. Department Result Summary
0. Exit
Enter choice:
Enter a valid whole number.
Enter choice: 8

--- DEPARTMENT RESULT SUMMARY ---
ECE | Enrolled course results: 1 | Pass count (approx): 1 | Fail count (approx): 0
IT | Enrolled course results: 3 | Pass count (approx): 0 | Fail count (approx): 3
MECH | Enrolled course results: 1 | Pass count (approx): 0 | Fail count (approx): 1
Note: Class average SGPA is not included in this compact summary; calculate per-student SGPA from the mark-sheet output.
```

DATABASE:

STUDENTS DATABASE:

```
mysql> SELECT * FROM students;
+----------+--------------+------------+------------------+
| roll_no  | student_name | department | current_semester |
+----------+--------------+------------+------------------+
| 23EER122 | Hari S       | ECE        |                7 |
| 23ITR156 | Arun k       | IT         |                7 |
| 23MER164 | Trishna K    | MECH       |                4 |
| 24ECR136 | Fathima I    | ECE        |                5 |
| 24MER178 | Rithika S    | MECH       |                5 |
| 25CSR132 | Kani J       | CSE        |                3 |
| 25CSR144 | Kumaran G    | CSE        |                3 |
| 25ITR121 | Sahana S     | IT         |                3 |
| 25ITR124 | SAMRUKA KK   | IT         |                3 |
| 25ITR144 | Subasree G   | IT         |                3 |
+----------+--------------+------------+------------------+
10 rows in set (0.00 sec)
```

COURSES DATABASE:

```
mysql> SELECT * FROM courses;
+-------------+--------------+---------+-------------+
| course_code | course_title | credits | course_type |
+-------------+--------------+---------+-------------+
| 22ITC32     | PC           |       3 | PRACTICAL   |
| 23ITP06     | DLD          |       2 | PRACTICAL   |
| 24ITC31     | ITC          |       4 | THEORY      |
| 24ITC45     | DESIGN LAB   |       2 | PRACTICAL   |
| 24ITP34     | JAVA         |       4 | THEORY      |
| 25ITC04     | EVS          |       1 | THEORY      |
| 25ITC31     | Full Stack   |       3 | THEORY      |
| 25ITP07     | PLDS         |       3 | EMBEDDED    |
+-------------+--------------+---------+-------------+
8 rows in set (0.00 sec)
```

ENROLLMENTS DATABASE:

```
mysql> SELECT * FROM enrollments;
+---------------+----------+-------------+----------+--------------+------------------+---------------------+
| enrollment_id | roll_no  | course_code | semester | classes_held | classes_attended | medical_condonation |
+---------------+----------+-------------+----------+--------------+------------------+---------------------+
|             2 | 25ITR144 | 25ITC31     |        3 |           60 |               30 |                   0 |
|             3 | 24ECR136 | 24ITP34     |        5 |            0 |                0 |                   0 |
|             5 | 24MER178 | 24ITP34     |        5 |           25 |               24 |                   0 |
|             6 | 25ITR124 | 24ITC45     |        3 |            0 |                0 |                   0 |
|             8 | 25ITR144 | 24ITC45     |        3 |            0 |                0 |                   0 |
+---------------+----------+-------------+----------+--------------+------------------+---------------------+
5 rows in set (0.00 sec)
```

ASSESSMENT MARKS DATABASE:

```
mysql> SELECT * FROM assessment_marks;
+---------------+-------+-------+-------------+---------+----------------+-----------+------------+
| enrollment_id | cia1  | cia2  | assignments | quizzes | lab_evaluation | ese_marks | ese_absent |
+---------------+-------+-------+-------------+---------+----------------+-----------+------------+
|             2 | 85.00 | 80.00 |       90.00 |    NULL |           NULL |     94.00 |          0 |
|             3 | 60.00 | 85.00 |       90.00 |    NULL |           NULL |     90.00 |          0 |
|             5 | 84.00 | 97.00 |       85.00 |    NULL |           NULL |      NULL |          1 |
|             6 |  NULL |  NULL |        NULL |    NULL |           NULL |      NULL |          0 |
|             8 |  NULL |  NULL |        NULL |    NULL |           NULL |      NULL |          0 |
+---------------+-------+-------+-------------+---------+----------------+-----------+------------+
5 rows in set (0.00 sec)
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

REFERENCES:

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GITHUB:

LINK:	https://github.com/Thulasi2508/java-project.git